

Statistics

Statistics is the collection, analysis, interpretation, and presentation of data.

What data could help a college decide when to keep its library open?

Descriptive statistics

Descriptive statistics summarizes the data we observed.

A survey finds that 36 of 60 students want later library hours. How could we summarize this result?

Inferential statistics

Inferential statistics uses sample evidence to draw conclusions about a population.

Can those 60 responses tell us what all students want?

Probability

Probability measures how likely an event is.

Does a 50% chance of heads guarantee five heads in ten tosses?

Long-run probability

A fair coin has probability 0.5 of heads on each independent toss.

Why should 4,000 tosses usually give a proportion nearer 0.5 than four tosses?

Population

The population is the entire group a study concerns.

To study this quarter's enrolled students, who belongs in the population?

Sample

A sample is the part of a population actually studied. Sampling is the process of selecting a sample.

Would surveying one afternoon class give every student a chance to be included?

Representative sample

A representative sample reflects the characteristics of the population relevant to the study.

Would a sample containing only morning classes represent all college students?

Variable

A variable is a characteristic recorded for each individual or object.

For a library-hours survey, what could we record for each student?

Numerical variables

A numerical (quantitative) variable records an amount with meaningful numerical units.

Is the number of library visits a numerical variable?

Categorical variables

A categorical (qualitative) variable assigns a label or category.

Why is preferred study location categorical even if locations receive number codes?

Classifying variables

A survey records each student's preferred study location and weekly library visits.

Which variable is categorical, and which is numerical?

Data

Data are the observed values of a variable.
One observed value is a datum.

For the variable “number of library visits last week,” what could three data values look like?

Parameter

A parameter is a numerical summary of a population.

What population percentage would the library-hours survey try to estimate?

Statistic

A statistic is a numerical summary of a sample.

Of 60 surveyed students, 36 want later hours. What sample statistic can we calculate?

Population or sample?

A college enrolls 2,400 students. A researcher randomly selects 80 of them for a survey.

Which group is the sample?

Parameter or statistic?

In a random sample of 80 students, 25% use the library on Fridays.

Is 25% a parameter or a statistic?

Variable or value?

A survey asks each student how many courses they take. One response is 4.

What is the variable, and what is the observed value?

Mean

The arithmetic mean, commonly called the average, is the sum of the values divided by their number.

Five study rooms contain 2, 3, 3, 5, and 7 students. What is the mean occupancy?

Proportion

A proportion is a part divided by its corresponding whole.

Of 40 reserved study rooms, 14 are unused. What proportion is unused?

A question of scope

A survey of 50 library visitors finds that 90% want longer library hours.

How far beyond these 50 visitors would you trust that result?

The terms in a mean study

Synthetic study: estimate mean pages printed per student at a college. Randomly survey 60 students; observed counts include 8, 12, and 19.

What are the population, sample, variable, data, parameter, and statistic?

The terms in a proportion study

Synthetic study: estimate the proportion of all campus printers needing repair. Inspect 20 randomly chosen printers; 3 need repair.

What are the population, sample, variable, data, parameter, and statistic?

Dot plots

A dot plot places one dot per observation above a number line, stacking repeated values. Synthetic visits: 0, 1, 1, 2, 2, 2, 4.

How would a dot plot show where these observations cluster?